

```

1  /* =====
2  3  BOOK_RECURRING_DEPOSIT
4  ===== */
5  CREATE OR ALTER PROCEDURE BookRecurringDeposit
6  (
7      @ACCOUNTID INT,
8      @DEPOSITAMOUNT DECIMAL(10,2),
9      @INTEREST DECIMAL(9,2),
10     @TENURE INT,
11     @CHANNEL VARCHAR(20) = 'Branch'
12 )
13 AS
14 BEGIN
15     SET NOCOUNT ON;
16
17     BEGIN TRY
18         BEGIN TRANSACTION;
19
20         -- 1. Validate deposit amount
21         IF @DEPOSITAMOUNT <= 0
22         BEGIN
23             THROW 50001, 'Deposit amount must be greater than zero.', 1;
24         END;
25
26         -- 2. Validate tenure
27         IF @TENURE <= 0
28         BEGIN
29             THROW 50002, 'Tenure must be greater than zero.', 1;
30         END;
31
32         -- 3. Validate interest rate
33         IF @INTEREST <= 0
34         BEGIN
35             THROW 50003, 'Interest rate must be greater than zero.', 1;
36         END;
37
38         -- 4. Validate channel
39         IF @CHANNEL NOT IN
40         (
41             'Branch',
42             'Internet Banking',
43             'Mobile Banking'
44         )
45         BEGIN
46             THROW 50004, 'Invalid channel for Recurring Deposit.', 1;
47         END;
48
49         -- 5. Validate account
50         IF NOT EXISTS
51         (
52             SELECT 1
53             FROM Accounts
54             WHERE AccountID = @ACCOUNTID
55                 AND AccountStatus = 'Active'
56         )
57         BEGIN
58             THROW 50005, 'Account does not exist or is not active.', 1;
59         END;
60
61         -- 6. Get account balance with locking
62         DECLARE
63             @OPENINGBALANCE DECIMAL(10,2),
64             @AVAILABLEBALANCE DECIMAL(10,2);
65
66         SELECT
67             @OPENINGBALANCE = CurrentBalance,
68             @AVAILABLEBALANCE = AvailableBalance
69         FROM Accounts WITH (UPLOCK, HOLDLOCK)
70         WHERE AccountID = @ACCOUNTID;
71
72         -- 7. Validate available balance
73         -- First RD installment must be available
74         IF @AVAILABLEBALANCE < @DEPOSITAMOUNT
75         BEGIN
76             THROW 50006, 'Insufficient available balance for Recurring Deposit.', 1;
77         END;
78
79         -- 8. Generate RD opening date
80         DECLARE @STARTDATE DATE;
81         SET @STARTDATE = CAST(SYSDATETIME() AS DATE);
82
83         -- 9. Calculate RD maturity date
84         DECLARE @ENDDATE DATE;
85         SET @ENDDATE = DATEADD(MONTH, @TENURE, @STARTDATE);
86
87         -- 10. Calculate maturity amount using function
88         DECLARE @MATURITYAMOUNT DECIMAL(10,2);
89         SET @MATURITYAMOUNT =
90             dbo.Calculate_RD_Maturity
91             (
92                 @DEPOSITAMOUNT,
93                 @INTEREST,
94                 @TENURE
95             );
96
97         -- 11. Update account balance
98         -- Deduct the first monthly installment
99         UPDATE Accounts
100         SET
101             CurrentBalance = CurrentBalance - @DEPOSITAMOUNT,
102             AvailableBalance = AvailableBalance - @DEPOSITAMOUNT,
103             LastTransactionDate = SYSDATETIME(),
104             ModifiedDate = SYSDATETIME()
105         WHERE AccountID = @ACCOUNTID;
106
107         -- 12. Insert Recurring Deposit record
108         INSERT INTO RecurringDeposits
109         (
110             AccountID,
111             MonthlyInstallment,
112             InterestRate,
113             TenureMonths,
114             OpeningDate,
115             MaturityDate,
116             MaturityAmount,
117             Status
118         )
119

```

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-- 13. Insert financial transaction
INSERT INTO Transactions
(
    TransactionReferenceNumber,
    TransactionReferenceNumber,
    AccountID,
    TransactionType,
    TransactionCategory,
    PaymentChannel,
    Amount,
    Charges,
    TaxAmount,
    OpeningBalance,
    ClosingBalance,
    CounterpartyAccountNumber,
    CounterpartyName,
    CounterpartyBank,
    TransactionStatus,
    Remarks,
    InitiatedBy,
    ApprovedBy,
    TransactionDate,
    CreatedDate
)
VALUES
(
    'RD-'
    + CONVERT(VARCHAR(8), CAST(SYSDATETIME() AS DATE), 112)
    + '-'
    + RIGHT('000000' + CAST(NEXT VALUE FOR TransactionReferenceSeq AS VARCHAR(6)), 6),
    NULL,
    @ACCOUNTID,
    'Debit',
    'RD Deposit',
    @CHANNEL,
    @DEPOSITAMOUNT,
    0.00,
    0.00,
    @OPENINGBALANCE,
    @OPENINGBALANCE - @DEPOSITAMOUNT,
    NULL,
    NULL,
    NULL,
    'Success',
    'Recurring Deposit booked via ' + @CHANNEL,
    NULL,
    NULL,
    SYSDATETIME(),
    SYSDATETIME()
);

-- 14. Commit complete RD booking
COMMIT TRANSACTION;

-- 15. Return booking result
SELECT
    @ACCOUNTID AS AccountID,
    @DEPOSITAMOUNT AS MonthlyInstallment,
    @INTEREST AS InterestRate,
    @TENURE AS TenureMonths,
    @STARTDATE AS OpeningDate,
    @ENDDATE AS MaturityDate,
    @MATURITYAMOUNT AS MaturityAmount,
    'Recurring Deposit booked successfully.' AS Message;

END TRY
BEGIN CATCH

    -- Rollback entire operation if any error occurs
    IF @@TRANCOUNT > 0
        ROLLBACK TRANSACTION;
    THROW;

END CATCH;

END;
GO

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@ACCOUNTID = 45, @DEPOSITAMOUNT = 5000, @INTEREST = 6.75,
@TENURE = 12, @CHANNEL = 'Branch';
--Example : RD through Mobile Banking
EXEC BookRecurringDeposit
@ACCOUNTID = 13, @DEPOSITAMOUNT = 10000, @INTEREST = 7.10,
@TENURE = 24, @CHANNEL = 'Mobile Banking';
-- Example : View updated account balances
select * from Accounts
where AccountID IN (39,13)
-- Example : View RD Booking transactions
select * from Transactions
where AccountID IN(39,13)
-- Example : View newly created Recurring Deposits
select * from RecurringDeposits
where AccountID IN(39,13)
-- Example : View audit records
select * from AuditLog
```

AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LienAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
13	BSGB002000013	16	2	6	2020-09-22	429167.87	429167.87	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-15 15:09:26.0892059	2026-08-09 17:07:30.1766834	2026-08-15 19:52:54.4482844
39	BSGB006000039	35	6	1	2024-02-16	217264.80	217264.80	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-06-13 00:00:00.0000000	2026-08-09 17:07:30.1766834	NULL

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remarks
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RDID	AccountID	MonthlyInstallment	InterestRate	TenureMonths	OpeningDate	MaturityDate	MaturityAmount	Status
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AuditID	TableName	RecordID	OperationType	OldValue	NewValue	ChangedBy	ChangedDate	IPAddress	EmployeeID
36	FixedDeposits	63	INSERT	NULL	FDNumber=FD-20260815-000042; AccountID=25; Amount=20000.00; Status=Active	SATYAMABHI	2026-08-15 12:43:33.8501385	NULL	NULL
37	FixedDeposits	64	INSERT	NULL	FDNumber=FD-20260815-000044; AccountID=47; Amount=39000.00; Status=Active	SATYAMABHI	2026-08-15 12:43:33.9488762	NULL	NULL

Before
RD Booking

Query executed successfully. SATYAM\SQLEXPRESS (16.0 RTM) SATYAM\ABHI (51) BankingSystem 00:00:00 4 rows

AccountID	MonthlyInstallment	InterestRate	TenureMonths	OpeningDate	MaturityDate	MaturityAmount	Message
39	5000.00	6.75	12	2026-08-15	2027-08-15	61891.50	Recurring Deposit booked successfully.
13	10000.00	7.10	24	2026-08-15	2028-08-15	257061.06	Recurring Deposit booked successfully.

After
RD Booking

AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LienAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
13	BSGB002000013	16	2	6	2020-09-22	409167.87	409167.87	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-15 19:52:54.4323286	2026-08-09 17:07:30.1766834	2026-08-15 19:52:54.4363503
39	BSGB006000039	35	6	1	2024-02-16	212264.80	212264.80	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-15 19:52:54.4323286	2026-08-09 17:07:30.1766834	2026-08-15 19:52:54.4492884

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remarks
48	RDB-20260815-000050	NULL	39	Debit	RD Deposit	Branch	5000.00	0.00	0.00	217264.80	212264.80	NULL	NULL	NULL	Success	Recurring Deposit
49	RDB-20260815-000051	NULL	13	Debit	RD Deposit	Mobile Banking	10000.00	0.00	0.00	419167.87	409167.87	NULL	NULL	NULL	Success	Recurring Deposit

RDID	AccountID	MonthlyInstallment	InterestRate	TenureMonths	OpeningDate	MaturityDate	MaturityAmount	Status
28	39	5000.00	6.75	12	2026-08-15	2027-08-15	61891.50	Active
29	13	10000.00	7.10	24	2026-08-15	2028-08-15	257061.06	Active

AuditID	TableName	RecordID	OperationType	OldValue	NewValue	ChangedBy	ChangedDate	IPAddress	EmployeeID
36	FixedDeposits	63	INSERT	NULL	FDNumber=FD-20260815-000042; AccountID=25; Amount=20000.00; Status=Active	SATYAMABHI	2026-08-15 12:43:33.8501385	NULL	NULL
37	FixedDeposits	64	INSERT	NULL	FDNumber=FD-20260815-000044; AccountID=47; Amount=39000.00; Status=Active	SATYAMABHI	2026-08-15 12:43:33.9488762	NULL	NULL
42	RecurringDeposits	28	INSERT	NULL	AccountID=39; MonthlyInstallment=5000.00; InterestRate=6.75; TenureMonths=12; MaturityAmount=61891.50; Status=Active	SATYAMABHI	2026-08-15 19:52:54.4363503	NULL	NULL
43	RecurringDeposits	29	INSERT	NULL	AccountID=13; MonthlyInstallment=10000.00; InterestRate=7.10; TenureMonths=24; MaturityAmount=257061.06; Status=Active	SATYAMABHI	2026-08-15 19:52:54.4492884	NULL	NULL

Query executed successfully. SATYAM\SQLEXPRESS (16.0 RTM) SATYAM\ABHI (51) BankingSystem 00:00:01 4 rows